SANKALCHAND PATEL UNIVERSITY

Visnagar

SWAMI SACHCHIDANAND POLYTECHNIC COLLEGE, VISNAGAR

A Report On

ONLINE VEHICAL RENTAL SYSTEM

D.E sem 6

(Information Teacanology)

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CERTIFICATE

This is to certify that **Goswami Krunal H.(2023095900018365), Barot Sumit M.(2023095900018357), Solanki Dilipkumar T.(2023095900018363)** Students of DE Semester 6 (Information and Communication Technology) has completed term work titled " **ONLINE VEHICAL RENTAL SYSTEM**" under my guidance in partial fulfillment of D.E. Computer Engineering (6th sem) Sankalchand patel university, Visnagar of during the academic year 2025-2026.

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ABSTARCT

- The Online Vehicle Rental System is a web-based platform designed to simplify the process of renting vehicles by providing users with a seamless and efficient booking experience. This system eliminates the need for traditional in person rentals, allowing customers to browse available vehicles, check pricing, make reservations, and complete payments online. The platform integrates key features such as real-time vehicle availability, user authentication, secure payment processing, and rental history tracking.

- The project aims to enhance customer convenience, optimize fleet management for rental providers, and ensure a secure and reliable transaction process. By leveraging modern web technologies, the system offers a user-friendly interface, automated booking confirmation, and improved business efficiency.

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CHAPTER 1

INTRODUCTON

1.1 Project Summary:

- The Online Vehicle Rental System is a web-based application designed to provide an easy and efficient way for customers to rent vehicles online. The main objective of this system is to simplify the traditional vehicle rental process and make it faster and more convenient for users. In the traditional system, customers need to visit the rental office to book a vehicle, but with this system, users can search and book vehicles through the internet.

- In this system, users can register and create their accounts. After logging in, users can view the list of available vehicles along with details such as vehicle name, type, price, and availability. Users can select a vehicle according to their needs and make an online booking.

- The system also includes an admin panel where the administrator can add, update, or delete vehicle details and manage customer bookings. All the information is stored in a database, which helps in easy data management and quick access to records.

- This project is developed using technologies such as **HTML, CSS, JavaScript, PHP, and MySQL**. The Online Vehicle Rental System helps improve the efficiency of vehicle rental services by providing a digital platform for both customers and administrators.

1.2 Purpose:

- The main purpose of the **Online Vehicle Rental System** is to provide an easy, fast, and convenient platform for users to rent vehicles online. This system helps customers to search, view, and book vehicles without visiting the rental office physically.

- Another purpose of this system is to reduce manual work and improve the efficiency of vehicle rental management. It allows the administrator to manage vehicle details, customer information, and bookings through an admin panel.

- This system also helps in maintaining records in a database, which makes data management easier and more organized. Overall, the purpose of this project is to make the vehicle rental process simple, time-saving, and more efficient for both customers and administrators.

1.3 Scope:

- The main modules of the projects are Customer Module, which performs all the operations related to customer such as adding new customer, edit the existing customer, search customer and delete customer. Car module, which performs all the operations related to car such as adding new car, edit the existing car, Details view car, search car and delete car.

1.4 Tecanology & Literature review:

1.4.1 Front-end Tecanology:

- Frontend refers to the part of the system that users interact with directly. In the Online Vehicle Rental System, frontend technologies are used to design the user interface of the website. Technologies such as **HTML, CSS, and JavaScript** are used to create web pages, forms, and navigation menus. These technologies help in making the website attractive, responsive, and easy to use for customers.

1.4.2 Back-end Tecanology:

- Backend refers to the server-side part of the application that processes user requests and performs system operations. In this project, **PHP** is used as the backend programming language. It handles tasks such as user registration, login, vehicle booking, and communication with the database.

1.4.3 Database Mangement:

- Database management is used to store and manage all the data of the system. In the Online Vehicle Rental System, **MySQL** is used as the database. It stores information such as user details, vehicle information, booking records, and admin data. The database helps in organizing and retrieving data efficiently.

1.4.4 Literature Review:

- The literature review focuses on studying existing vehicle rental systems and related technologies. Many traditional vehicle rental services use manual processes for booking and managing vehicles, which can be time-consuming and less efficient. Modern web-based systems allow users to book vehicles online, check availability, and manage bookings easily. The Online Vehicle Rental System is developed to provide a digital platform that improves efficiency, reduces manual work, and offers better service to customers.

CHAPTER 2

SYSTEM REQUIREMENT STUDY

2.1 Use Characteristics:

- The users of the Online Vehicle Rental System include customers and administrators. Customers are the main users who use the system to search, view, and book vehicles online. They are expected to have basic knowledge of using computers or smartphones and internet browsing. The system is designed with a simple and user-friendly interface so that even users with limited technical knowledge can easily access and use the system. Administrators manage the system by adding vehicles, updating vehicle details, managing bookings, and monitoring user activities. They are expected to have moderate technical knowledge and basic understanding of system management. The system can be used occasionally or frequently depending on the user's need for renting vehicles.

2.2 Hardware And Software Requirement:

2.2.1 Hardware Requirment:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Hard Disk:** Minimum 500 GB
- **Monitor:** Standard LCD/LED Monitor
- **Keyboard and Mouse:** For user input
- **Internet Connection:** Required for accessing the web application
- **Server System:** For hosting the website and database

2.2.2 Software Requirments:

- **Operating System:** Windows 10 / Windows 11
- **Front-End Technology:** HTML, CSS, JavaScript
- **Back-End Technology:** PHP
- **Database:** MySQL
- **Development Tool:** Visual Studio Code / Notepad++
- **Web Server:** XAMPP / Apache Server

2.3 Constraints:

- The Online Vehicle Rental System has some constraints. First, a stable **internet connection** is required because the system works online. Second, **data security** is important to protect the personal information of users. Third, the system must be **compatible with different devices** such as laptops and mobile phones. In addition, **time and budget limitations** may also affect the development of the project.

2.3.1 Regulatory Policies:

- Regulatory policies are the rules and guidelines that the system must follow. In an Online Vehicle Rental System, the company must follow **government transport rules**, **valid driving license verification**, and **user data protection policies**. The system should also follow **payment and privacy regulations** to keep customer information safe. These policies help ensure that the service is legal, secure, and trustworthy for users.

2.3.2 Hardware Limitation:

- Hardware limitation means the system may depend on the available hardware devices. If the computer, server, or network hardware is slow or outdated, the system performance can be affected. Limited **RAM, storage, or processing power** may reduce the speed of the system. Therefore, proper hardware is required for the smooth working of the Online Vehicle Rental System.

2.3.3 Interface to other application:

- The Online Vehicle Rental System may need to connect with other applications to work properly. It can interface with **payment gateway applications** for online payments, **map or GPS services** for vehicle location, and **email or SMS services** to send booking confirmations and notifications to users. These interfaces help the system provide better and faster services.

2.3.4 Parallel operations:

- Parallel operations mean that the system can perform multiple tasks at the same time. In the Online Vehicle Rental System, many users can **search vehicles, make bookings, and process payments simultaneously**. The system should handle these operations efficiently without affecting performance or causing delays. This ensures smooth and fast service for all users.

2.3.5 Reliability Requirements:

- Reliability requirements ensure that the Online Vehicle Rental System works correctly and consistently. The system should provide accurate vehicle information, booking details, and payment processing. It must work properly without frequent errors or crashes. The system should also store data safely and recover information in case of system failure to maintain reliable service for users.

2.3.6 Criticality of the application:

- The Online Vehicle Rental System is an important application because it manages vehicle booking, customer information, and payment details. If the system fails, users may not be able to book vehicles or access their booking information. Therefore, the system must work accurately, securely, and without interruption to ensure smooth service for customers and administrators.

2.3.7 Safety and security consideration:

- Safety and security are very important in the Online Vehicle Rental System. The system must protect **user personal information, login details, and payment data** from unauthorized access. Proper **authentication and password protection** should be used to keep the system secure. The system should also maintain **safe data storage and backup** to prevent data loss and ensure reliable service.

2.3.8 Application and Dependencies:

- The Online Vehicle Rental System depends on several applications and technologies to work properly. It requires a **web browser**, **internet connection**, and a **server** to run the application. The system also depends on **database management systems**, **payment gateway services**, and **email or SMS services** for notifications. These dependencies help the system function smoothly and provide better service to users.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Study of current system:

- The current system for vehicle rental is mostly manual. Customers need to visit the rental office to check vehicle availability and complete the booking process. All details such as customer information, vehicle records, and payment details are recorded manually. This process takes more time, requires more paperwork, and there are chances of errors or data loss. Therefore, an online system is needed to make the process faster, easier, and more efficient.

3.2 Problem of weakness of current system:

- The current vehicle rental system has several weaknesses. The process is mostly **manual**, which takes more time and effort. Customers must **visit the rental office** to check vehicle availability and make bookings. There is also a high chance of **human errors** in maintaining records. Managing customer and vehicle data on paper can lead to **data loss or mismanagement**. In addition, it is difficult to **track vehicle availability and bookings quickly**, which reduces efficiency.

3.3 Requirement of New system:

- A new Online Vehicle Rental System is required to make the rental process faster and more efficient. The system should allow users to **view available vehicles, register, and book vehicles online**. It should store **customer details, vehicle information, and booking records** in a database. The system should also provide **secure login, online payment options, and booking confirmation**. This new system will reduce manual work, save time, and improve service for both customers and administrators.

3.4 Feasibility study:

- A feasibility study checks whether the Online Vehicle Rental System can be successfully developed and implemented. It includes:

  - **Technical Feasibility** – The system can be built using current technology like web development tools, databases, and payment gateway integration.

  - **Economic Feasibility** – The cost of development, hardware, and maintenance is reasonable compared to the benefits of automation and improved service.

- **Operational Feasibility** – Users and staff can easily use the system for booking, managing vehicles, and processing payments.

- **Legal Feasibility** – The system will comply with government transport rules, data privacy laws, and online payment regulations.

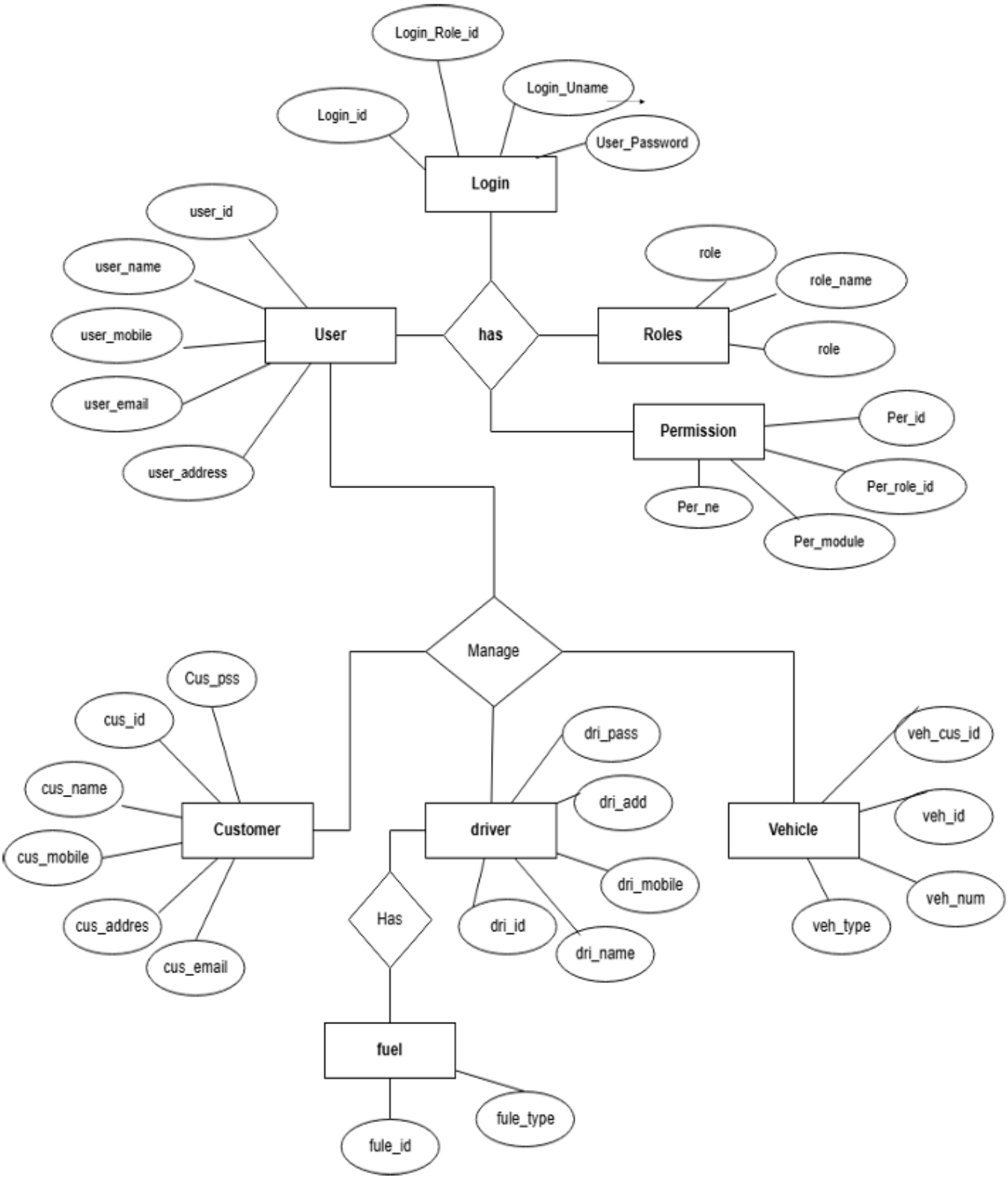
3.5 Requirements Validation:

- Requirements validation ensures that all the system requirements are **complete, correct, and feasible** before development begins. For the Online Vehicle Rental System, this means checking that features like **user registration, vehicle booking, payment processing, and data storage** meet the needs of both users and administrators. It also ensures that the system will work **securely, reliably, and efficiently**. Validation helps prevent errors, reduces development cost, and ensures the final system fulfills its intended purpose.

CHAPTER 4

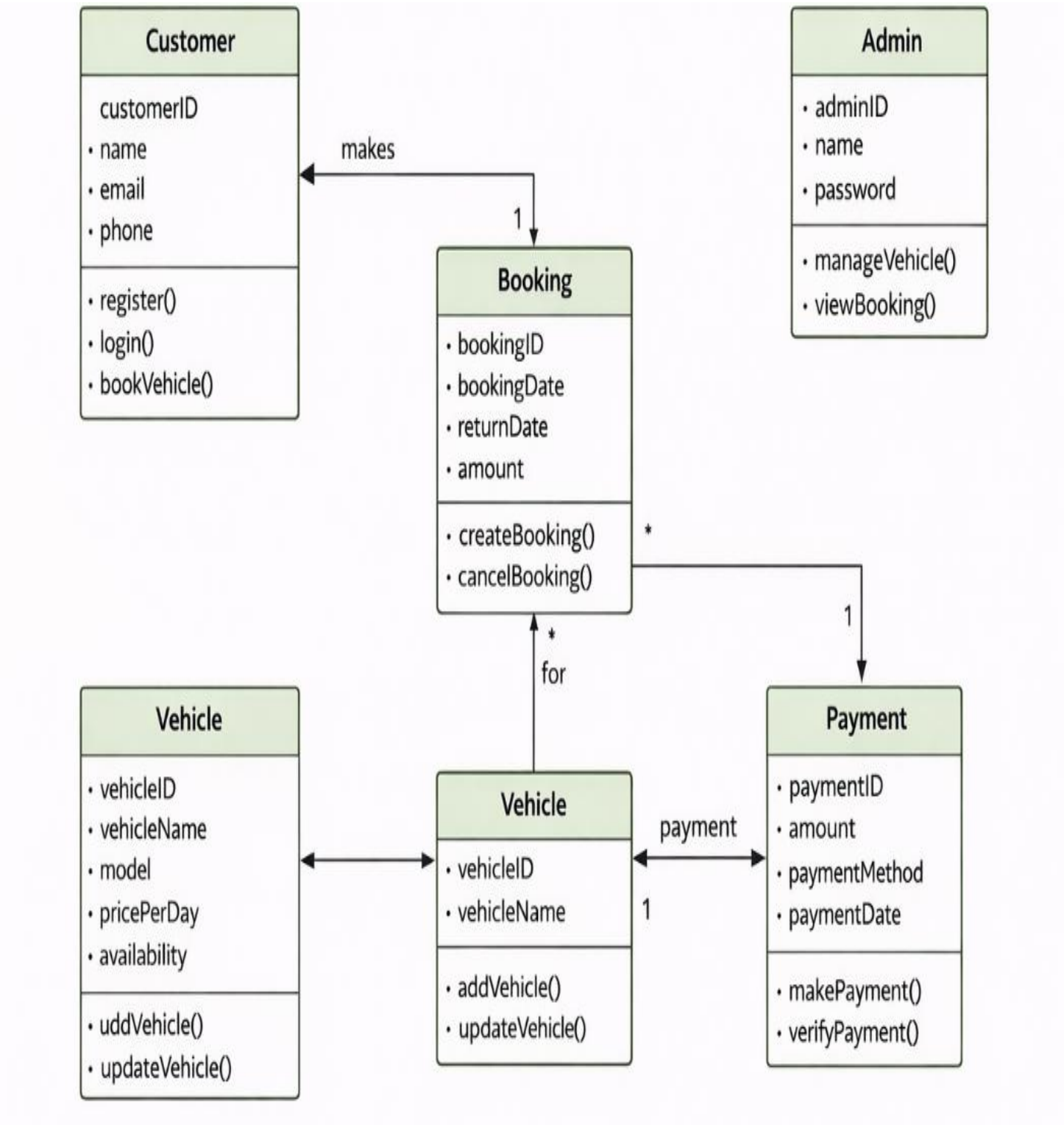
DATA MODELING

4.1 E-R Diagram:



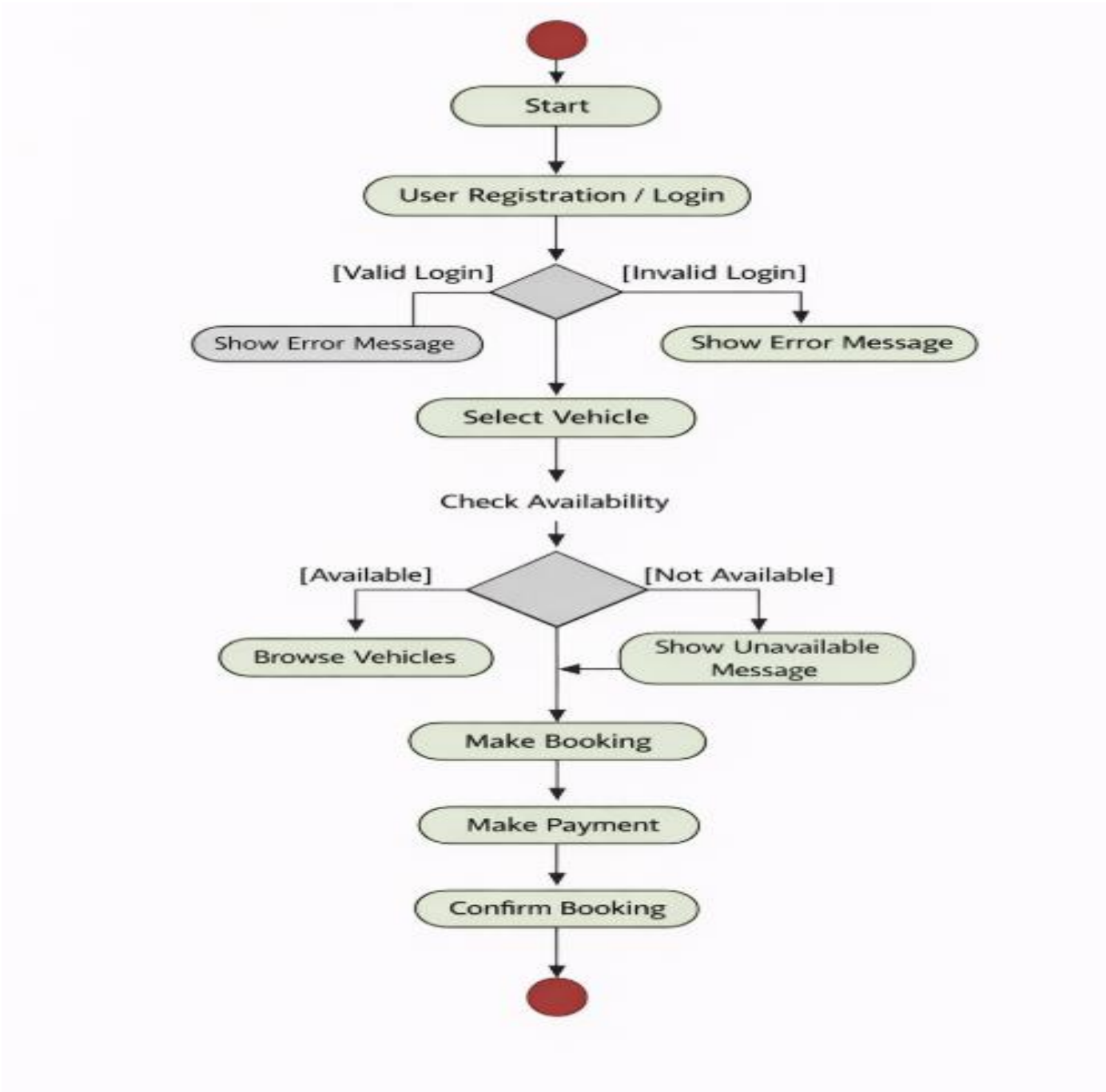
(4.1 E-R Diagram)

4.2 Class Diagram:



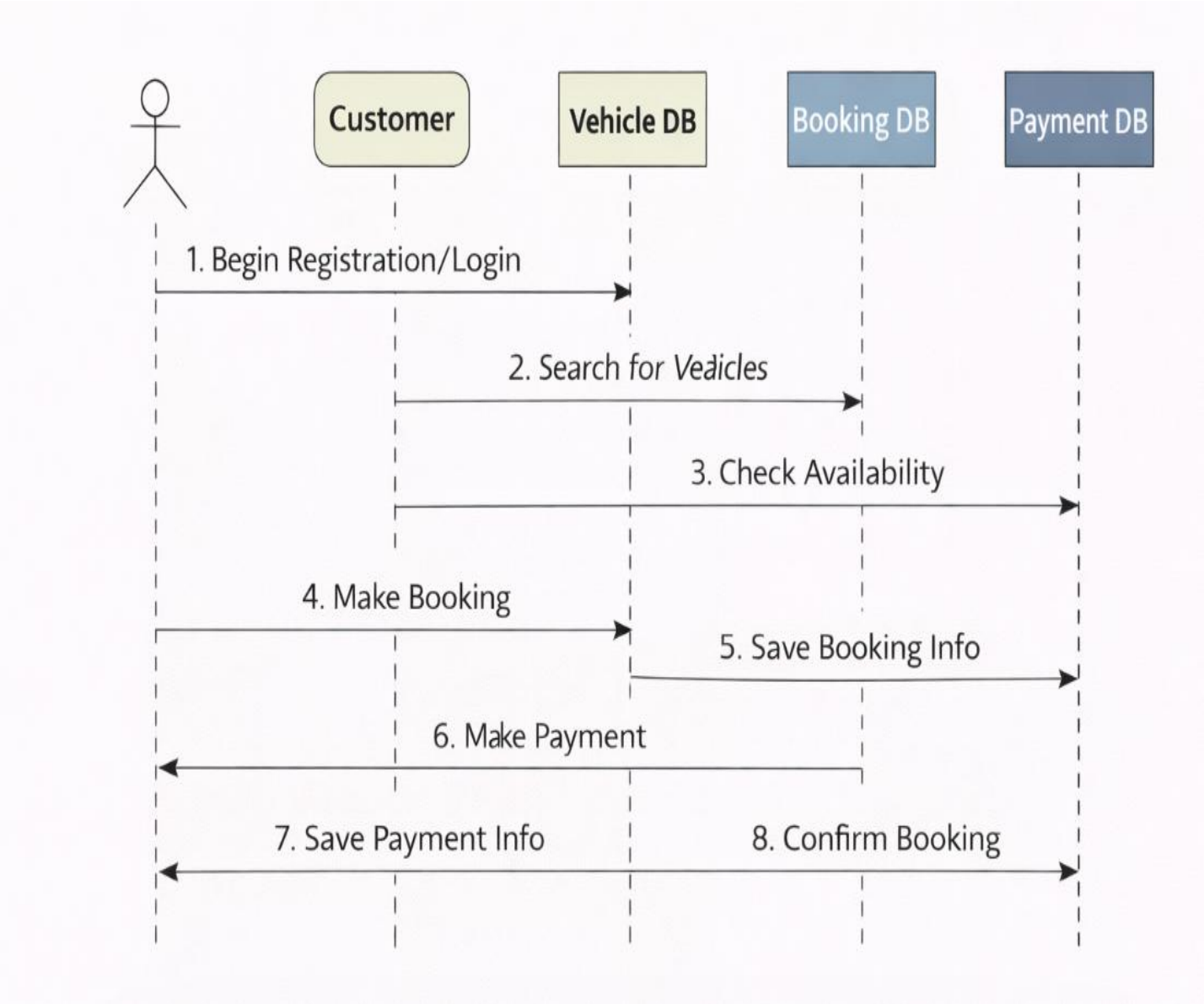
(4.2 Class Diagram)

4.3 Activity Diagram:



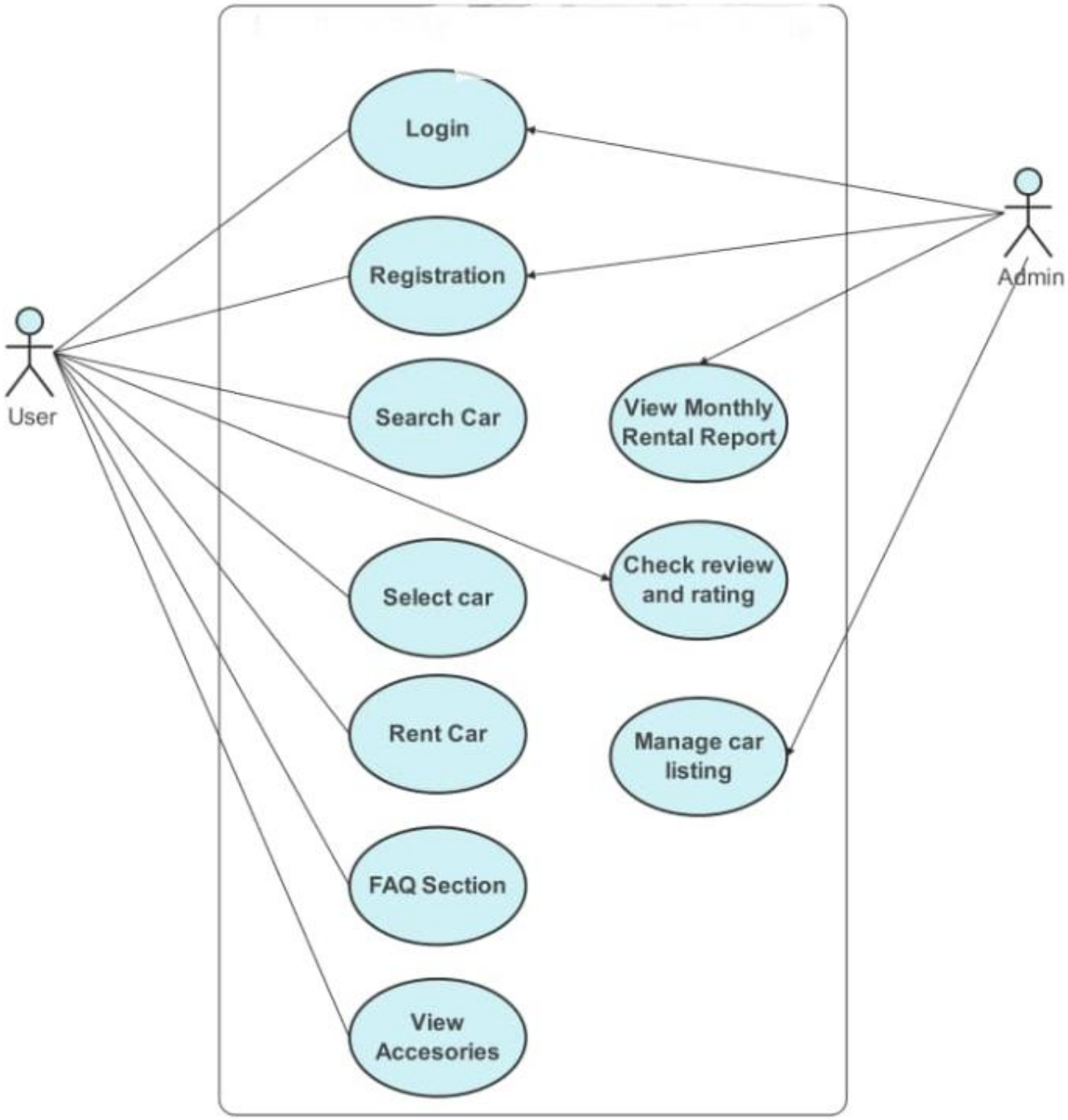
(4.3 Activity Diagram)

4.4 Sequence Diagram:



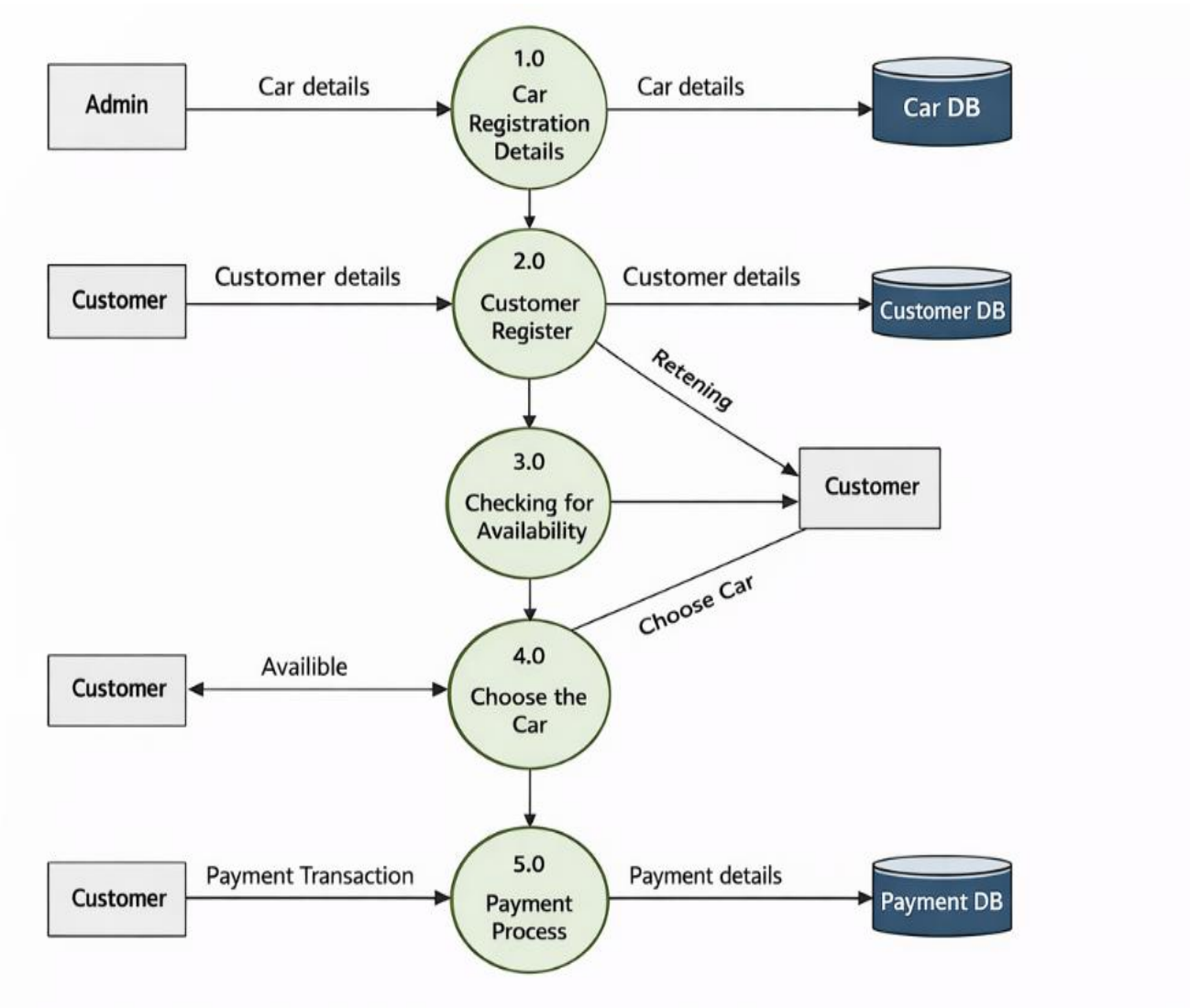
(4.4 Sequence Diagram)

4.5 Use Case Diagram:



(4.5 Use Case Diagram)

4.6 Data Flow Diagram:



(4.6 Data Flow Diagram**)**

CHAPTER 5

DATABASE DESIGN

5.1Data Dictionary:

Table Name: User

Column	Type	Description	Constraint
id	Int	User Id	PRIMARY KEY
FullName	varchar	Full Name	NULL
EmailId	varchar	Email Id	NULL
Password	varchar	Password	NULL
ContactNo	char	Contact No	NULL
dob	varchar	Date Of Birth	NULL
Address	varchar	Address	NULL
City	varchar	City	NULL
Country	varchar	Country	NULL

5.2. Table Name: Admin

Column	Type	Description	Constraint
id	int	Admin id	PRIMARY KEY
UserName	varchar	User Name	NULL
Password	varchar	Password	NULL

5.3 Table Name: TBL – Booking:

Column	Type	Description	Constraint
id	int	Booking Id	PRIMARY KEY
UserEmail	varchar	Email	NULL
Vehicle-Id	int	Vehicle Id	NULL
From-Date	date	From Date	NULL
To-Date	date	To Date	NULL
message	varchar	Message	NULL
Status	int	Status	NULL

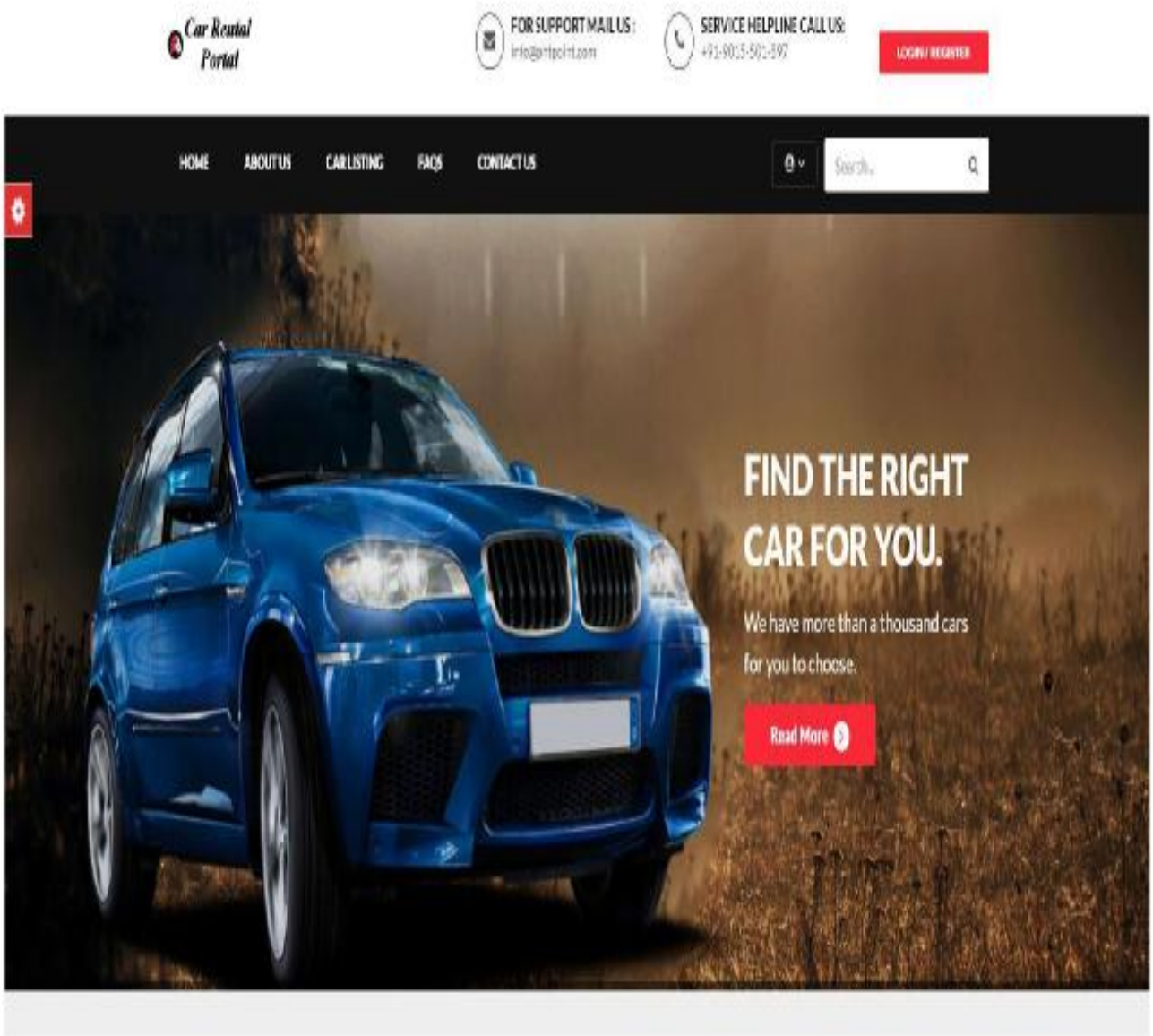
5.4. Table Name: TBL-Brands:

Column	Type	Description	Constraint
Id	int	Contact Id	PRIMARY KEY
Address	Varchar	Address	NULL
EmailId	varchar	Email id	NULL
ContactNo	char	Contact number	NULL

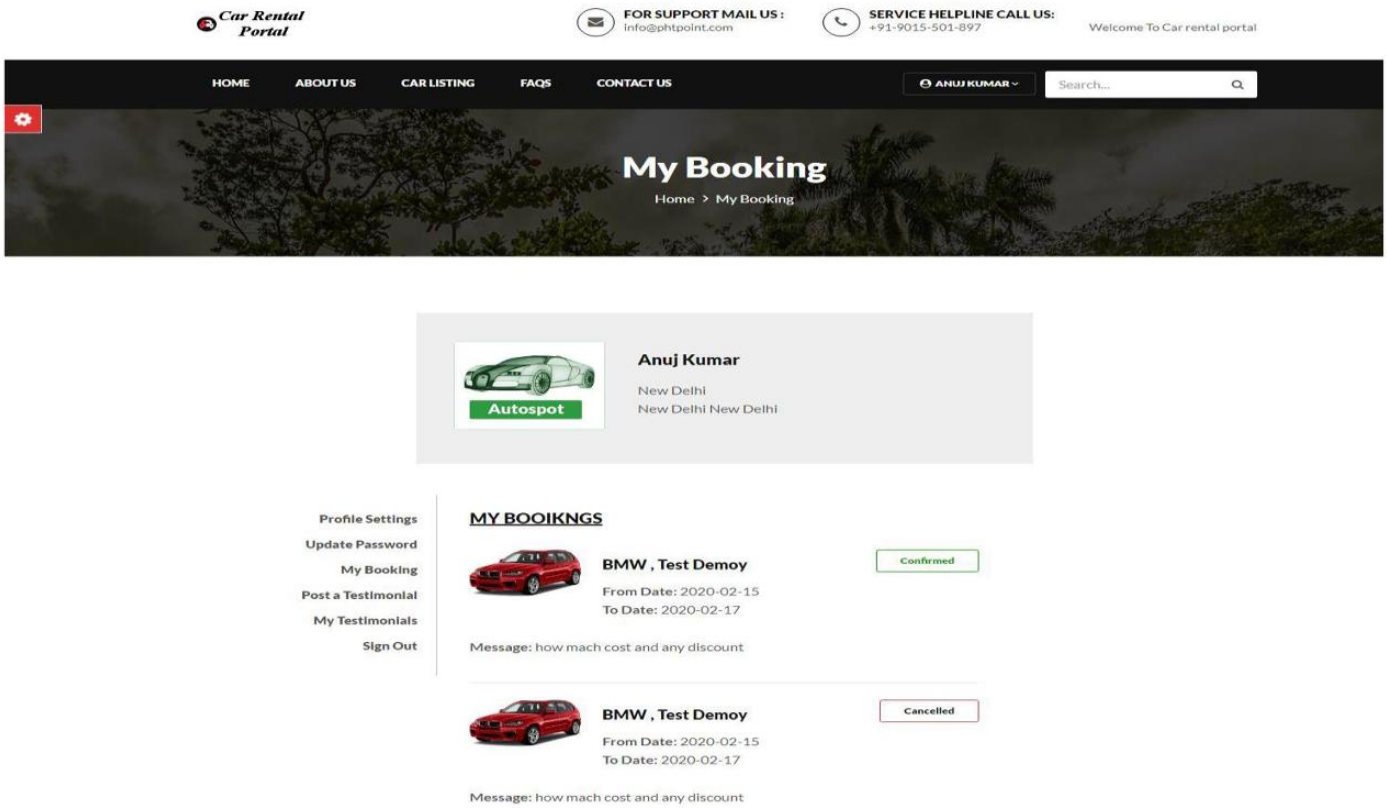
CHAPTER 6

DESIGN

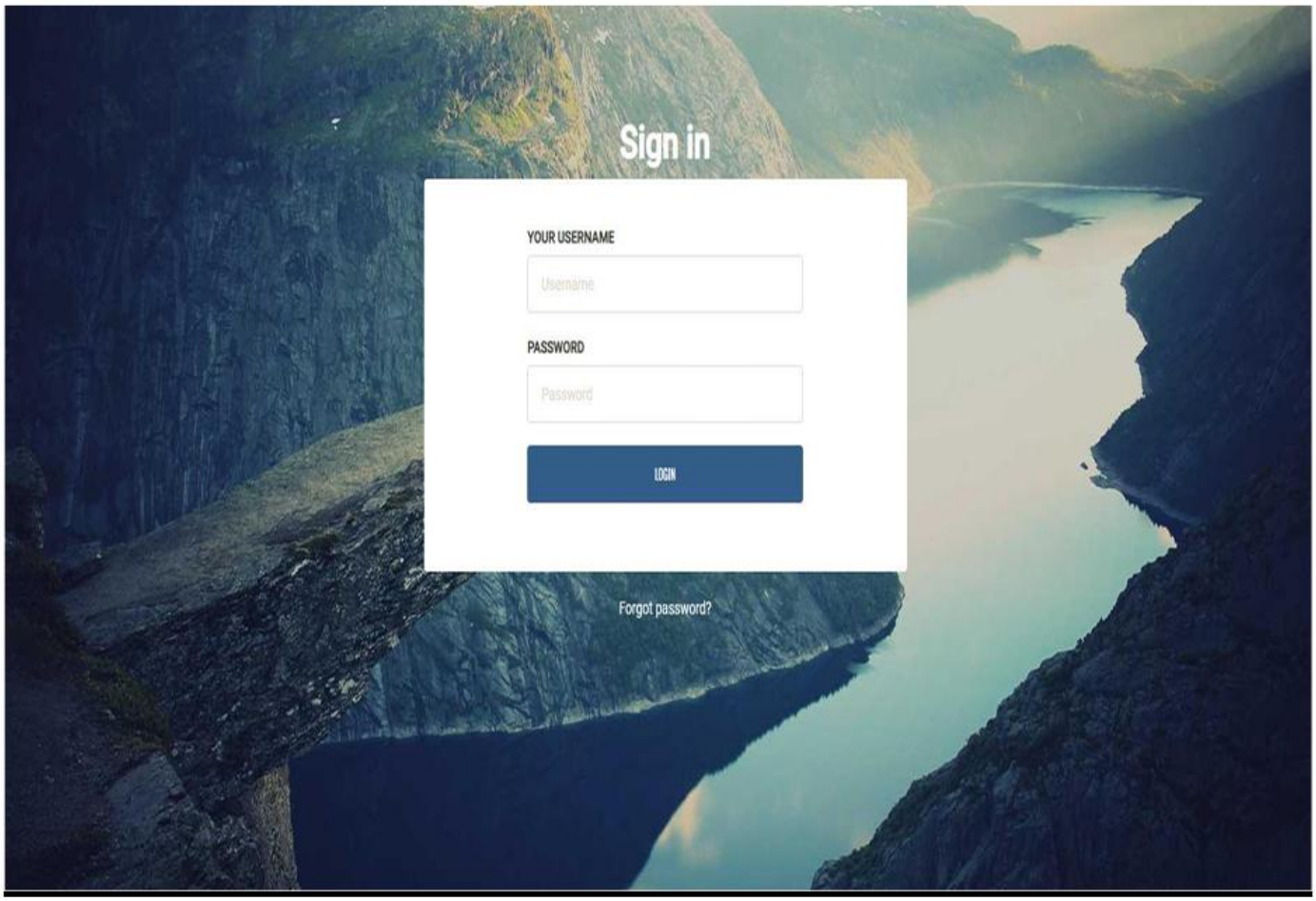
6.1 User Page:


Car Rental Portal
FOR SUPPORT MAIL US :
SERVICE HELPLINE CALL US:
HOME
ABOUT US
CAR LISTING
FAQS
CONTACT US
FIND THE RIGHT CAR FOR YOU.
We have more than a thousand cars for you to choose.
Read More

6.2 Your Booking:



6.3 Admin Login:



CHAPTER 7

TESTING

7.1 Introduction Of Testing:

- Software testing is the process of checking a software system to find errors and ensure that it works according to the given requirements. It helps to improve the quality, reliability, and performance of the software before it is delivered to the users. Testing ensures that the system functions correctly and meets user needs.

7.2 Testing Strategy:

- esting strategy is a plan used to test the software system to ensure it works correctly and meets the requirements. It defines the testing methods, techniques, and types of testing such as unit testing, integration testing, system testing, and acceptance testing. The main purpose of a testing strategy is to find errors and ensure the quality and reliability of the software system.

7.3 Testing Methods:

- Testing methods are used to check whether the **Online Vehicle Rental System** works properly and meets user requirements. Methods like **Black Box Testing** are used to test booking, login, and payment functions without seeing the internal code. **White Box Testing** checks the internal program logic, and **Grey Box Testing** checks both the system functions and internal structure to ensure the system works correctly and efficiently.

❖ Conclusion:

- The Online Vehicle Rental System provides a seamless, efficient, and user friendly platform for renting vehicles online. By leveraging digital solutions, the system enhances accessibility, reduces manual efforts, and streamlines the rental process for both customers and service providers.

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